

Irregular Waves

Course : CE60222

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Wind-Wave Generation and Decay

- Wind blowing over the surface of a water body will transfer energy to the water in the form of a surface current and by generating waves on the water surface.
- This process is best explained by a resonance model proposed by Phillips (1957,1960).
- There are turbulent eddies in the wind field that exert a fluctuating pressure on the water surface. These pressure fluctuations vary in magnitude and frequency and they move forward at a range of speeds.
- The pressure fluctuations cause water surface undulations to develop and grow.
- The key to their growth is that a resonant interaction occurs between the forward moving pressure fluctuations and the free waves that propagate at the same speed as the pressure fluctuations.

Wind-Wave Generation and Decay

- Although the Phillips model explains the initiation of wave motion, it is insufficient to explain the continued growth of the waves.
- This growth is best explained by a shear flow model proposed by Miles (1957).
- As the wind blows over a forward moving wave a complex air flow pattern develops over the wave.
- This involves a secondary air circulation that is set up around an axis that is parallel to the wave crest, by the wind velocity profile acting over a moving wave surface profile.
- Below a point on the velocity profile where the wind velocity equals the wave celerity, air flow is reversed relative to the forward moving wave profile.

Wind-Wave Generation and Decay

- Above this point air flow is in the direction of the wave motion.
- This results in a relative flow circulation in a vertical plane above the wave surface that causes a pressure distribution on the surface that is out of phase with the surface displacement.
- The result is a momentum transfer to the wave that selectively amplifies the steeper waves.
- It is desirable to select a single wave height and period to represent a spectrum of wind-waves for use in wave prediction, wave climate analysis, design of coastal structures, and so on.
- If the wave heights from a wave record are ordered by size one can define a height H_n that is the average of the highest n percent of the wave heights.

Wind-Wave Generation and Decay

- For example, H_{10} is the average height of the highest 10% of the waves in the record and H_{100} is the average wave height.
- The most commonly used representative wave height is H_{33} , which is the average height of the highest one-third of the waves.
- H_{33} is commonly called the significant height H_s and it is approximately the height an experienced observer will report when visually estimating the height of waves at sea.
- These highest waves tend to have periods that are around the middle of the range of periods in a wave spectrum.

Wind-Wave Generation and Decay

- So we most commonly define a significant period T_s as the average period of the highest one-third of the waves in the record.
- The significant wave height and period as well as the resulting spectrum of wind-generated waves depend primarily on the distance over which the wind blows (known as the fetch length F), the wind velocity W (commonly measured at the 10 m elevation), and the duration of the wind t_d .
- Other factors that affect the resulting waves generated by a wind field are the fetch width, the water depth and bottom characteristics if the depth is sufficiently shallow, atmospheric stability, and the temporal and spatial variations in the wind field during wave generation.
- Waves are generated with propagation directions aligned at a range of oblique angles ($< 90^\circ$) to the direction of the wind.

Wind-Wave Generation and Decay

- The range of directions decreases with an increase in wave period as waves grow while propagating along the fetch.
- Thus, the smaller the fetch width the lesser the chance shorter waves have of remaining in the generating area and growing to appreciable size.
- The water depth affects the wave surface profile form and water particle kinematics and thus the transfer of energy from the wind to the waves.
- Bottom friction dissipates wave energy and thus retards the rate of wave growth and the ultimate wave size.

Wind-Wave Generation and Decay

- It is instructive to consider the growth of the significant height and period as a function of distance along a fetch for waves generated by a wind of constant velocity, blowing over a constant fetch and having different durations.
- If the wind duration exceeds the time required for waves to travel the entire fetch length, i.e. $t_d > F/C_g$, the waves will grow to OAB as shown in and their characteristics at the end of the fetch will depend on F (for given W). This is the fetch limited condition
- If the wind duration is less ($t_d < F/C_g$), wave growth stops at OAC ($x = F_{\min}$) and wave generation is duration limited.
- If both the fetch and duration are sufficiently large, the curve OAB becomes essentially horizontal at the downwind end and a fully developed sea (FDS) has been generated for that particular wind speed.

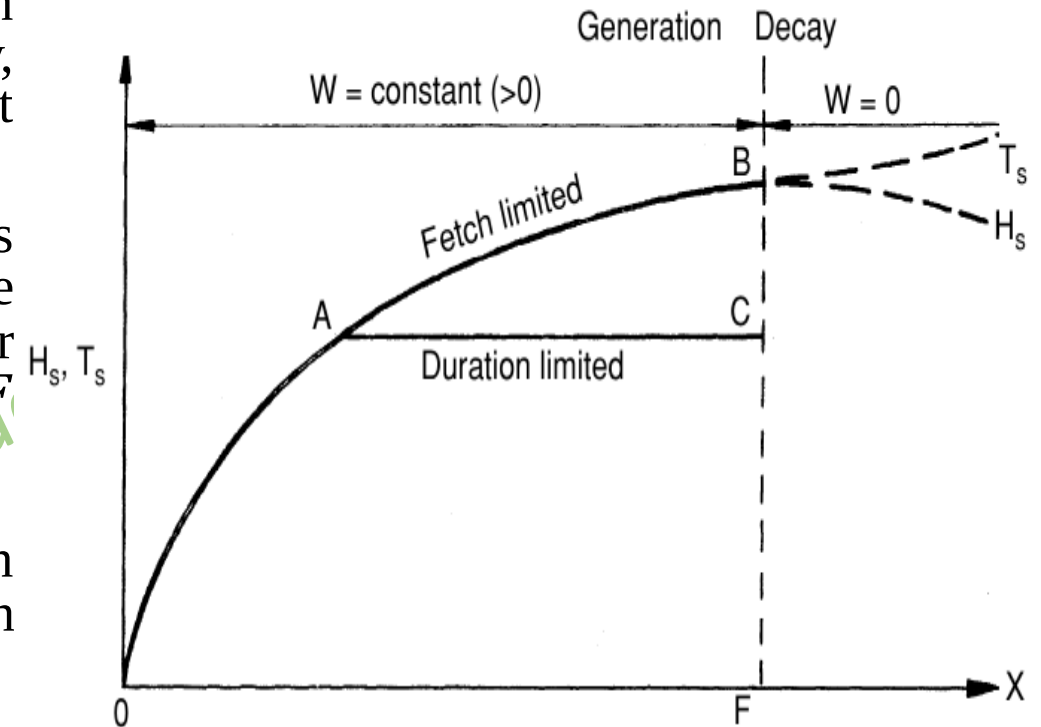


Figure 1. Idealized wave growth and decay for a constant wind velocity.

Wind-Wave Generation and Decay

- Outside of the region where the wind is blowing the waves propagate as swell.
- In this region the significant height will decrease and the significant period will increase.
- Energy dissipation and lateral spreading of the waves will decrease the wave height.
- This effect is greater for the shorter period waves so the significant period will increase.

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Wind-Wave Generation and Decay

- The characteristics of the waves generated by a given wind condition may also be defined by a wave spectrum.
- This is a plot of the wave energy density at each component period or frequency versus the range of component periods or frequencies.
- Figure 2 shows a series of typical wave spectra at successive points along a fetch. Note the decrease in the peak frequency (increase in peak period) as the wave spectrum grows along the fetch.
- The total area under the spectral curve, which is related to the spectral energy and significant wave height, also grows.

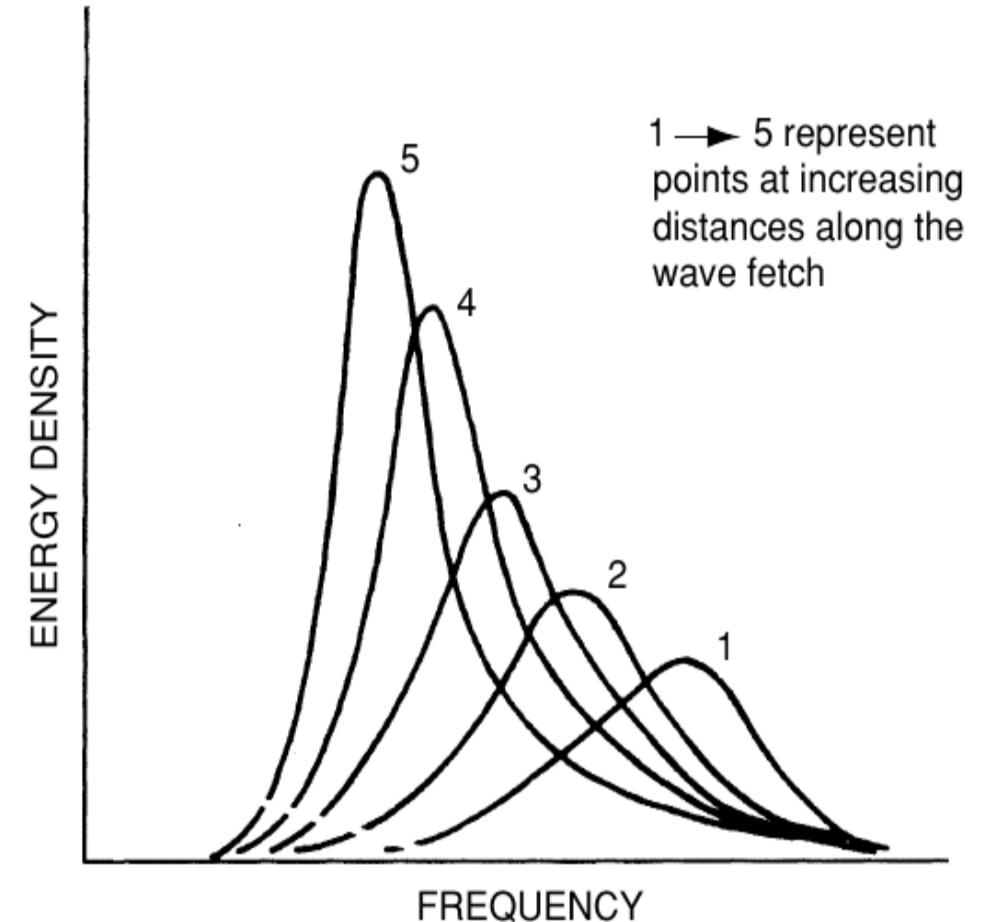


Figure 2 Wave spectra growth.

Wind-Wave Generation and Decay

- The higher frequency (lower period) waves on the right side of a spectrum grow to an energy level or wave height that is limited by breaking, so as further growth occurs it must take place at the lower frequencies (higher periods).
- Also, wave-wave interactions transfer some wave energy from lower to higher wave periods as the spectrum grows.
- If the wave spectra were measured at a fixed point in the wind field as time elapses, the spectra would exhibit a time-dependent growth that is similar to the growth pattern depicted in Figure 2.

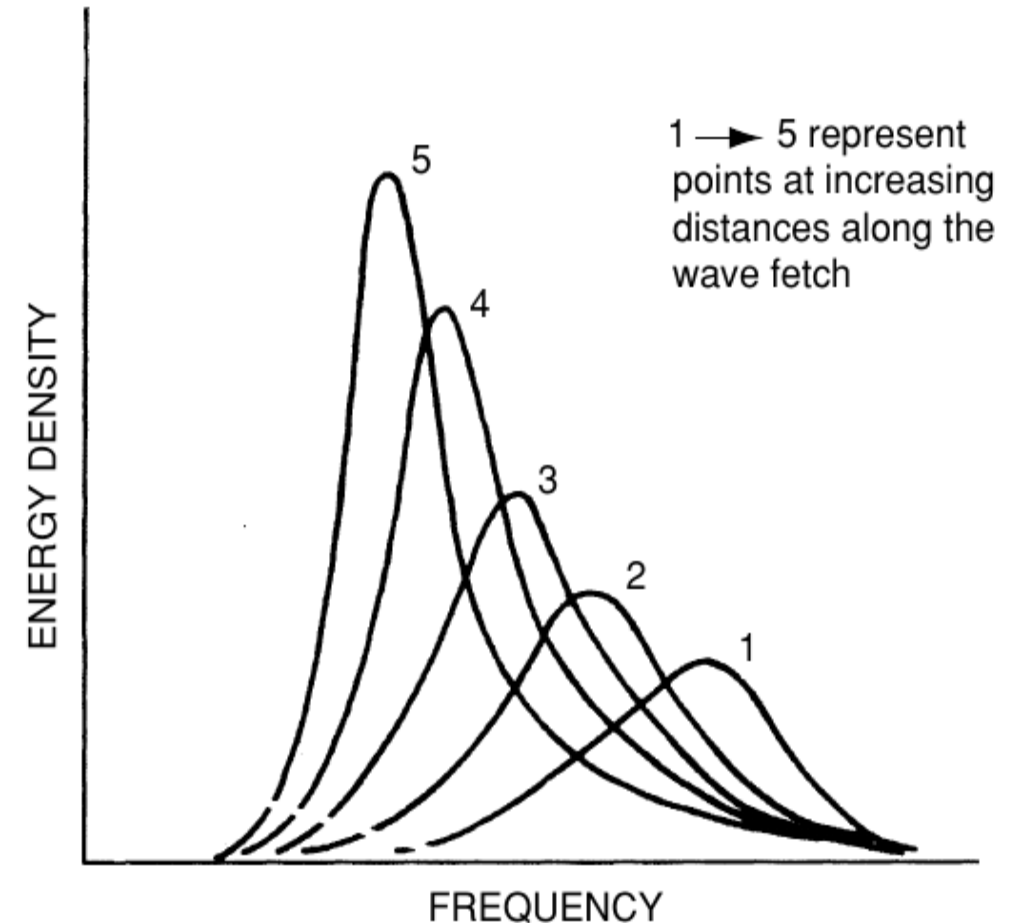


Figure 2 Wave spectra growth.

Wave Record Analysis for Height and Period

Wave Height Distribution

- The most commonly used analysis procedure is the zero-up crossing method (Pierson, 1954).
- A mean water surface elevation is determined and each point where the water surface crosses this mean elevation in the upward direction is noted.
- Figure 3 shows a short segment of a typical wave record.
- The analysis procedure must be statistically reasonable and consistent.
- The time elapsed between consecutive points is a wave period and the maximum vertical distance between crest and trough is a wave height.

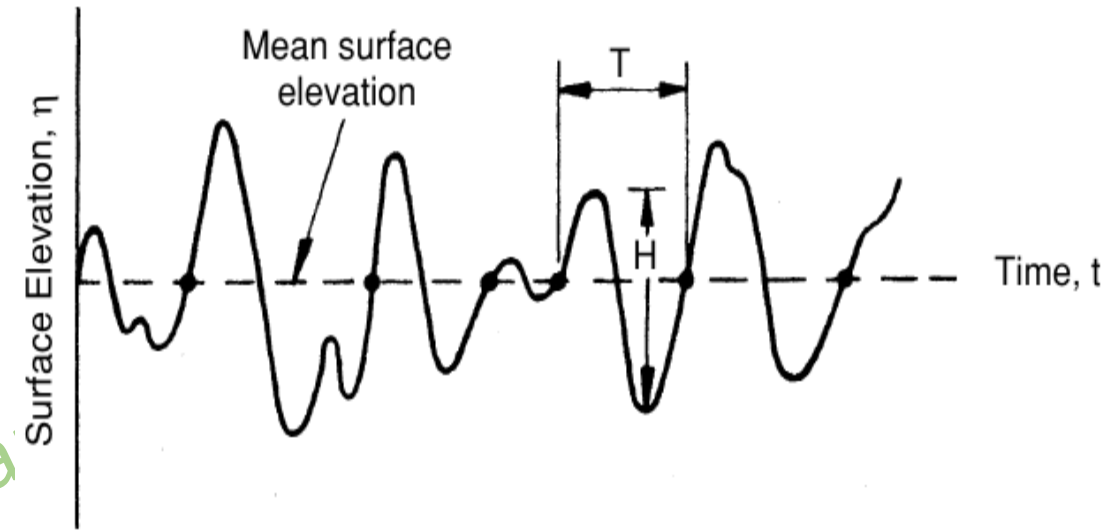


Figure 3. Typical water surface elevation versus time record.

- Note that some small surface undulations are not counted as waves so that some higher frequency components in the wave record are filtered out.
- Focus is primarily on the larger waves in the spectrum. This is not of major concern, since for engineering purposes our

Wave Record Analysis for Height and Period

- For engineering purposes it is desirable to have a model for the distribution of wave heights generated by a storm.
- Longuet-Higgins (1952) demonstrated that this distribution is best defined by a Rayleigh probability distribution.
- Use of this distribution requires that the wave spectrum has a single narrow band of frequencies and that the individual waves are randomly distributed.
- Practically, this requires that the waves be from a single storm that preferably is some distance away so that frequency dispersion narrows the band of frequencies recorded.

Wave Record Analysis for Height and Period

- The Rayleigh distribution can be written

$$p(H) = \frac{2H}{(H_{\text{rms}})^2} e^{-(H/H_{\text{rms}})^2}$$

where the root mean square height H_{rms} is given by

$$H_{\text{rms}} = \sqrt{\sum \frac{H_i^2}{N}}$$

H_i are the individual wave heights in a record containing N waves. Employing the Rayleigh distribution leads to the following useful relationships:

$$H_s = 1.416H_{\text{rms}}$$
$$H_{100} = 0.886H_{\text{rms}}$$

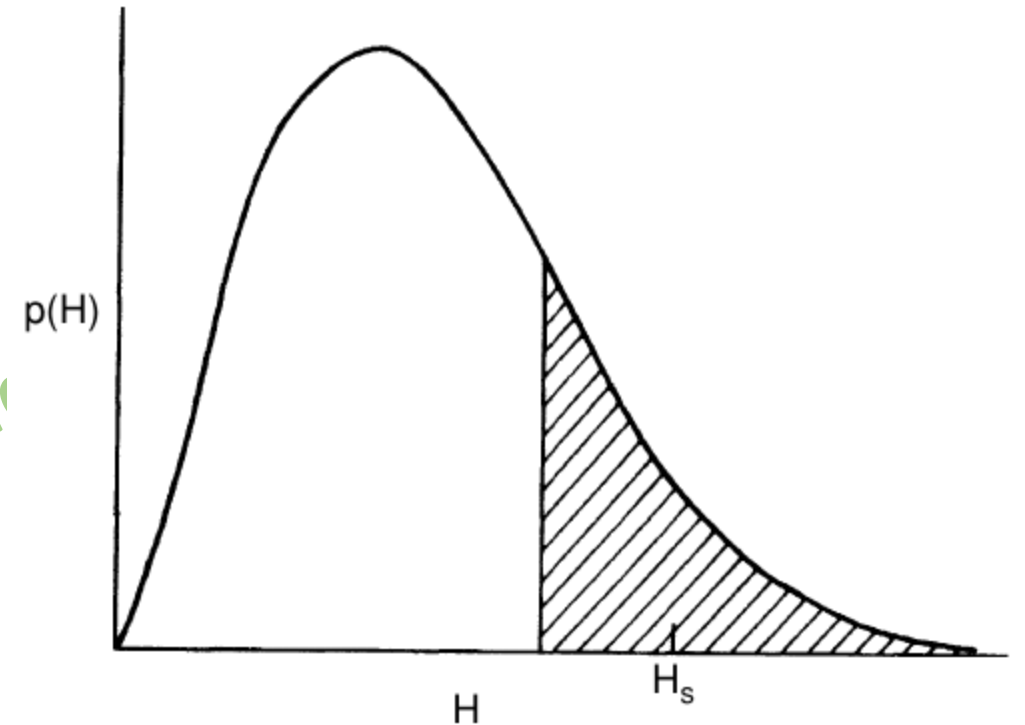


Figure 4. Typical wave height–frequency distribution.

Wave Record Analysis for Height and Period

- The cumulative probability distribution $P(H)$ (i.e., the percentage of waves having a height that is equal to or less than H) is

$$P(H) = \int_0^H p(H)dH = 1 - e^{-(H/H_{\text{rms}})^2}$$

- For our purposes, we are more interested in the percentage of waves that have a height greater than a given height, i.e.,

$$1 - P(H) = e^{-(H/H_{\text{rms}})^2}$$

Since $H_s = 1.416H_{\text{rms}}$

$$1 - P(H_s) = e^{-(1.416)^2} = 0.135$$

- So 13.5% of the waves in a storm wave record might have heights that are greater than the significant height.

Wave Record Analysis for Height and Period

- Figure 5, which is adapted from the U.S. Army Coastal Engineering Research Center (1984), is a plot that is useful when applying the Rayleigh distribution.
- Line *a* in the figure gives the probability P that any wave height will exceed the height (H/H_{rms}) and line *b* gives the average height of the n highest fraction of the waves.

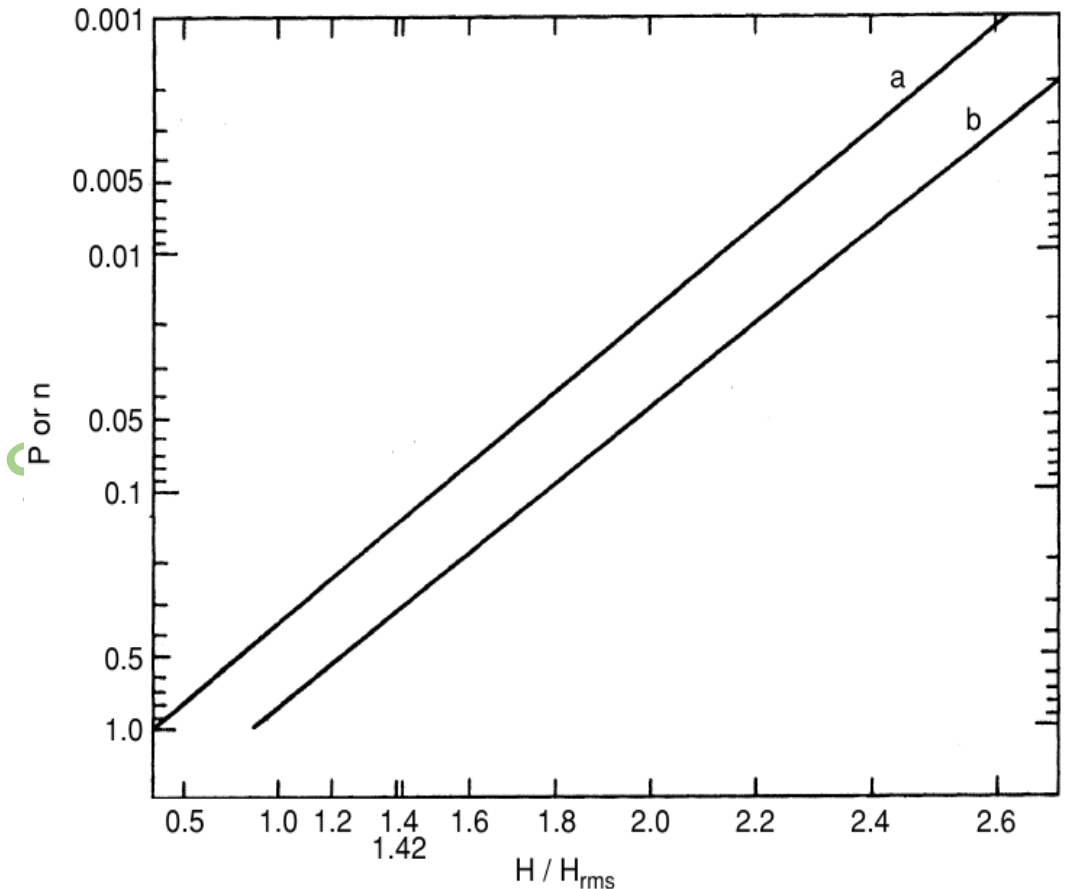


Figure 5. Raleigh distribution for wave heights.

Problem

A wave record taken during a storm is analyzed by the zero-up crossing method and contains 205 waves. The average wave height in the record is 1.72 m. Estimate H_s , H_5 , and the number of waves in the record that would exceed 2.5 m height.

Solution,

$$H_s = 1.416H_{rms}$$

$$H_{100} = 0.886H_{rms}$$

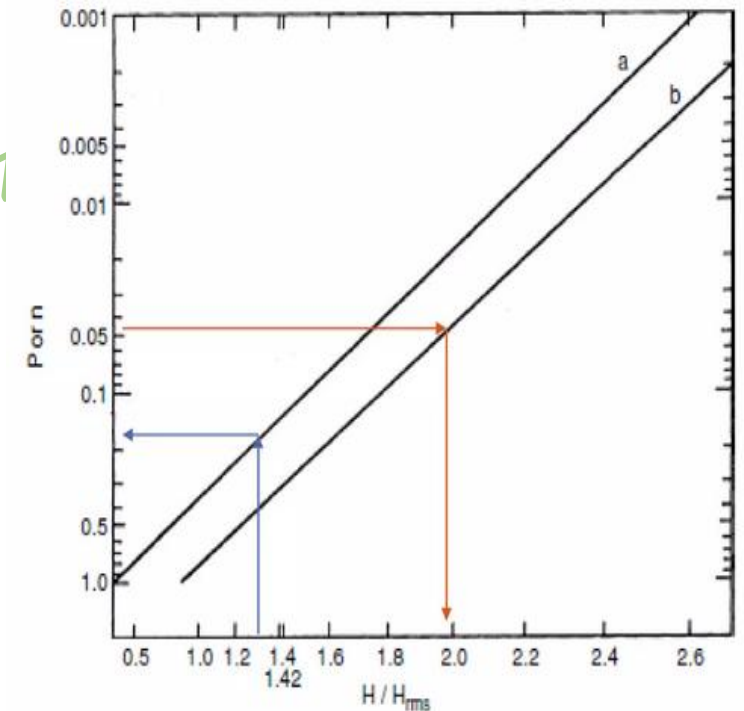
$$H_{rms} = \frac{1.72}{0.886} = 1.94 \text{ m}$$

and

$$H_s = 1.416(1.94) = 2.75 \text{ m}$$

From line b in $H_5/H_{rms} = 1.98$ so

$$H_5 = 1.98(1.94) = 3.84 \text{ m}$$



Problem

From line a

$$\frac{H}{H_{\text{rms}}} = \frac{2.5}{1.94} = 1.29$$

we have

$$P = 0.19.$$

So, the estimated number of waves exceeding 2.5 m in height is $205(0.19) = 38.95$, or approximately 39 waves

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Maximum Wave Height

- There is no upper limit to the wave heights defined by the Rayleigh distribution.
- In a storm, however, the highest wave that might be expected will depend on the length of the storm as well as its strength. Longuet-Higgins (1952) demonstrated that for a storm with a relatively large number of waves N , the expected value of the height of the highest wave $H_{\max}$ would be

$$H_{\max} = 0.707H_s\sqrt{\ln N}$$

- For example, a storm having a 6 hour duration of high waves having an average period of 8 s would about 2700 waves and $H_{\max}$ would be $1.99H_s$.
- In the nearshore zone, the highest wave might be limited by wave breaking, provided the storm can generate sufficiently high waves for this limit to apply in the water depth of concern.
- However, in deeper water beyond the depth where waves would break, this equation should be appropriate.

Characteristics wave height

- Average wave height (H or H_{avg})
- Root mean square wave height (H_{rms})
- Significant wave height (H_s or $H_{1/3}$)
- One-tenth wave height ($H_{1/10}$)
- Maximum wave height (H_{max})

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Characteristics wave height

- **Average wave height (H or H_{avg})**

$$H_{avg} = \frac{1}{N} \sum_{j=1}^N H_j$$

Where, N = number of waves; H_j = individual wave height

- **Root mean square height (H_{rms})**

$$H_{rms} = \sqrt{\frac{1}{N} \sum_{j=1}^N H_j^2}$$

Characteristics wave height

- **Significant wave height (H_s or $H_{1/3}$)**

This is average wave height of the largest 33% of the waves.

- **One-tenth wave height ($H_{1/10}$)**

This is average wave height of the largest 10% of the waves.

- **Maximum wave height (H_{max})**

This maximum wave height in a wave train.

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Problem

For a wave train containing 51 waves and corresponding periods are given in Table. Determine the characteristics wave heights i.e.

1. Average wave height (H or H_{avg})
2. Root mean square wave height (H_{rms}),
3. Significant wave height (H_s or $H_{1/3}$),
4. One-tenth wave height ($H_{1/10}$),
5. Maximum wave height (H_{max}).

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Problem

Sr. No.	Wave height (m)	Wave period (s)	Sr. No.	Wave height (m)	Wave period (s)	Sr. No.	Wave height (m)	Wave period (s)
1	0.6	5.2	18	2.8	8.2	35	1.18	10
2	1.92	8.5	19	2.75	5.9	36	0.96	9.7
3	4.41	7.3	20	2.76	4.3	37	1.22	6.7
4	2.64	12.4	21	3.02	8.9	38	0.88	8
5	3.1	8.3	22	3.88	9.4	39	2.45	9.3
6	1.95	5.8	23	2.07	8.9	40	3.86	5.7
7	1.9	6.4	24	1.99	7.6	41	4.66	11.3
8	0.85	7.2	25	3.51	6.7	42	1.46	8.7
9	2	4.3	26	2.47	5.5	43	3.32	9.1
10	2.14	8.3	27	1.93	6.1	44	4.36	5.6
11	1.28	5.1	28	0.49	5.8	45	5.11	8.3
12	2.07	8.7	29	2.35	6.9	46	4.89	9.1
13	1.89	6.6	30	1.96	8.1	47	2.76	10
14	1.75	7.9	31	3.45	10.8	48	3.55	12.4
15	4.11	8.8	32	2.96	9.4	49	1.06	11.8
16	5.4	8.2	33	3.61	8.6	50	0.93	9.4
17	2.45	9.4	34	2.38	9.0	51	154	8.4

Steps:

- Compute H^2 .
- Sort wave height H in descending order.
- Make a column containing 33% of the highest waves for estimation of significant wave height.
- Make another column similarly for the highest one-tenth waves.
- Estimate H_{rms} using $H_{rms} = \sqrt{\frac{1}{N} \sum_{j=1}^N H_j^2}$.
- Find the means of H , H_s and $H_{1/10}$ columns.
- Report H_{max} as the maximum wave height in the record provided.

Sr. No.	Wave height (<i>m</i>)	Wave period (<i>s</i>)	H^2	Wave height in descending order	H_s or $H_{1/3}$	$H_{1/10}$
1	0.6	5.2	0.4	5.4	5.4	5.4
2	1.92	8.5	3.7	5.11	5.11	5.11
3	4.41	7.3	19.4	4.89	4.89	4.89
4	2.64	12.4	7.0	4.66	4.66	4.66
5	3.1	8.3	9.6	4.41	4.41	4.41
6	1.95	5.8	3.8	4.36	4.36	
7	1.9	6.4	3.6	4.11	4.11	
8	0.85	7.2	0.7	3.88	3.88	
9	2	4.3	4.0	3.86	3.86	
10	2.14	8.3	4.6	3.61	3.61	
11	1.28	5.1	1.6	3.55	3.55	
12	2.07	8.7	4.3	3.51	3.51	
13	1.89	6.6	3.6	3.45	3.45	
14	1.75	7.9	3.1	3.22	3.22	
15	4.11	8.8	16.9	3.1	3.1	
16	5.4	8.2	29.2	3.02	3.02	
17	2.55	9.4	6.5	2.96	2.96	
18	2.8	8.2	7.8	2.8		

Sr. No.	Wave height (<i>m</i>)	Wave period (<i>s</i>)	H^2	Wave height in descending order	H_s or $H_{1/3}$	$H_{1/10}$
19	2.75	5.9	7.6	2.76		
20	2.76	4.3	7.6	2.76		
21	3.02	8.9	9.1	2.75		
22	3.88	9.4	15.1	2.64		
23	2.07	8.9	4.3	2.55		
24	1.99	7.6	4.0	2.47		
25	3.51	6.7	12.3	2.45		
26	2.47	5.5	6.1	2.38		
27	1.93	6.1	3.7	2.35		
28	0.49	5.8	0.2	2.14		
29	2.35	6.9	5.5	2.07		
30	1.96	8.1	3.8	2.07		
31	3.45	10.8	11.9	2		
32	2.96	9.4	8.8	1.99		
33	3.61	8.6	13.0	1.96		
34	2.38	9.6	5.7	1.95		
35	1.18	10	1.4	1.93		

Sr. No.	Wave height (m)	Wave period (s)	H^2	Wave height in descending order	H_s or $H_{1/3}$	$H_{1/10}$
36	0.96	9.7	0.9	1.92		
37	1.22	6.7	1.5	1.9		
38	0.88	8	0.8	1.89		
39	2.45	9.3	6.0	1.75		
40	3.86	5.7	14.9	1.54		
41	4.66	11.3	21.7	1.46		
42	1.46	8.7	2.1	1.28		
43	3.22	9.1	10.4	1.22		
44	4.36	5.6	19.0	1.18		
45	5.11	8.3	26.1	1.06		
46	4.89	9.1	23.9	0.96		
47	2.76	10	7.6	0.93		
48	3.55	12.4	12.6	0.88		
49	1.06	11.8	1.1	0.85		
50	0.93	9.4	0.9	0.6		
51	1.54	8.7	2.4	0.49		
	$H_{avg} = 2.53$	$T_{avg} = 8.10$	$H_{rms} = 7.88$		$H_s = 3.95$	$H_{1/10} = 4.89$
	$H_{max} = 5.4$					

Wave Spectral Characteristics

- An alternate approach to analyzing a wave record is by determining the resulting wave spectrum for that record.
- A water surface elevation time history can be reconstructed by adding a large number of component sine waves that have different periods, amplitudes, phase positions, and propagation directions.
- A directional wave spectrum is produced when the sum of the energy density in these component waves at each wave frequency $S(f, \theta)$ is plotted versus wave frequency f and direction θ .
- Commonly, one-dimensional wave spectra are developed when the energy for all directions at a particular frequency $S(f)$ is plotted as a function of only wave frequency.

Wave Spectral Characteristics

- An alternate form to the above described frequency spectrum is the period spectrum where the wave energy density $S(T)$ is plotted versus the wave period.
- From the small-amplitude wave theory, the energy density in a wave is $\rho g H^2 / 8$. Leaving out the product of the fluid density and the acceleration of gravity, as is commonly done, leads to the following expression for a directional wave spectrum:

$$S(f, \theta) df d\theta = \sum_f^{f+df} \sum_{\theta}^{\theta+d\theta} \frac{H^2}{8}$$

where H is the height of the component waves making up the spectrum. This simplifies to

$$S(f) df = \sum_f^{f+df} \frac{H^2}{8}$$

Wave Spectral Characteristics

- For a one-dimensional frequency spectrum. For a one-dimensional period spectrum we have

$$S(T)dT = \sum_T^{T+dT} \frac{H^2}{8}$$

- It can be shown (see Sorensen, 1993) that the following relationship holds:

$$S(f) = S(T)T^2$$

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Wave Spectral Characteristics

- A general form of a spectral model equation is

$$S(f) = \frac{A}{f^5} e^{-B/f^4}$$

where A and B adjust the shape and scale of the spectrum and can be written either as a function of the generating factors or as a function of a representative wave height and period (e.g., H_s and T_s).

- An important way to characterize a wave spectrum is by the moments of a spectrum. The n th moment of a spectrum is defined as

$$m_n = \int_0^{\infty} S(f) f^n df$$

- So, for example, the zeroth moment would just be the area under the spectral curve.
- Since a spectrum plot shows the energy density at each frequency versus the range of frequencies, the area under the spectral curve is equal to the total energy density of the wave spectrum (divided by the product of the fluid density and acceleration of gravity).

Wave Spectral Characteristics

- The spectral moment concept is useful to define a representative wave height. From the small-amplitude wave theory, the total energy density is twice the potential energy density of a wave. Thus,

$$\bar{E} = 2\bar{E}_p = \frac{2}{T_*} \int_0^{T_*} \rho g \eta \left(\frac{\eta}{2} \right) dt$$

where T_* is the length of wave record being analyzed and the overbar denotes energy density. This can be written

$$\bar{E} = \rho g \bar{\eta}^2 = \frac{\rho g \sum \eta^2}{N_*}$$

where the overbar denotes the average of the sum of N_* digitized water surface elevation values from a wave record of length T_* . From our definition of H_{rms} and H_s the energy density can also be written

$$\bar{E} = \frac{\rho g H_{\text{rms}}^2}{8} = \frac{\rho g H_s^2}{16}$$

Wave Spectral Characteristics

- If the zeroth moment of the spectrum equals the energy density divided by ρg we have,

$$\bar{E} = \rho g m_o = \rho g \frac{\sum \eta^2}{N_*}$$

and

$$\bar{E} = \rho g m_o = \frac{\rho g H_s^2}{16}$$

- Thus significant height definition from the wave spectrum energy density or zeroth moment:

$$H_s = 4\sqrt{m_o}$$

where the designation H_{m_o} will be used for this definition of significant height.

- Recall that H_s is based on a wave-by-wave analysis from the wave record, but H_{m_o} is determined from the energy spectrum or, more basically, from digitized values of the water surface elevation given by the wave record.

Wave Spectral Models

- Four of these spectral models-called the Bretschneider, Pierson-Moskowitz, JONSWAP, and TMA spectra-will be presented.
- These models are of interest from an historic perspective and because of their common use in coastal engineering practice.
- The first three models were developed for deep water waves and the last is adjusted for the effects of water depth.

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Wave Spectral Models

Bretschneider Spectrum (Bretschneider, 1959)

- The basic form of this spectrum is

$$S(T) = \frac{\alpha g^2}{(2\pi)^4} T^3 e^{-0.675(gT/2\pi W F_2)^4}$$

where W is the wind speed at the 10 m elevation and

$$\alpha = 3.44 \frac{F_1^2}{F_2^2}$$
$$F_1 = \frac{gH_{100}}{W^2} \quad F_2 = \frac{gT_{100}}{2\pi W}$$

- H_{100} and T_{100} denote the average wave height and period. The parameters F_1 and F_2 are a dimensionless wave height and dimensionless wave period, respectively.

Wave Spectral Models

Pierson-Moskowitz Spectrum (Pierson and Moskowitz, 1964)

- They selected records representing essentially fully developed seas for wind speeds between 20 and 40 knots to produce the following spectrum:

$$S(f) = \frac{\alpha g^2}{(2\pi)^4 f^5} e^{-0.74(g/2\pi W f)^4}$$

- The wind speed W is measured at an elevation of 19.5 m which yields a speed that is typically 5% to 10% higher than the speed measured at the standard elevation of 10 m.
- The coefficient α has a value of 8.1×10^{-3} .

Wave Spectral Models

Pierson-Moskowitz Spectrum (Pierson and Moskowitz, 1964)

- The following relationships can be developed from the Pierson-Moskowitz spectrum formulation (see Ochi, 1982):

$$H_{mo} = \frac{0.21W^2}{g}$$
$$f_p = \frac{0.87g}{2\pi W}$$

- The simple form of the Pierson-Moskowitz spectrum results in it being used in some situations where the sea is not fully developed.

Wave Spectral Models

JONSWAP spectrum

- Pierson-Moskowitz spectrum can be written as

$$S(f) = \frac{\alpha g^2}{(2\pi)^4 f^5} e^{-1.25(f_p/f)^4}$$

- The JONSWAP spectral form is a modification of Pierson-Moskowitz spectrum by developing relationships for α and f_p in terms of the wind speed and fetch, and enhancing the peak of the spectrum by a factor γ . The resulting spectrum is

$$S(f) = \frac{\alpha g^2}{(2\pi)^4 f^5} e^{-1.25(f_p/f)^4} \gamma^a$$

where

$$a = e^{-[(f-f_p)^2 / 2\sigma^2 f_p^2]}$$

$$\sigma = 0.07 \text{ when } f < f_p$$

$$\sigma = 0.09 \text{ when } f \geq f_p$$

Wave Spectral Models

- In the JONSWAP spectrum, γ typically has values ranging from 1.6 to 6 but the value of 3.3 is recommended for general usage.
- The coefficient γ is simply the ratio of $S(f)$ at the peak frequency for the JONSWAP and Pierson-Moskowitz spectra. This is depicted in Figure 6, which demonstrates the effect of γ on the spectrum shape.
- The coefficient α and the peak frequency f_p for the JONSWAP spectrum are given by

$$\alpha = 0.076 \left(\frac{gF}{W^2} \right)^{-0.22}$$

$$f_p = \frac{3.5g}{W} \left(\frac{gF}{W^2} \right)^{-0.33}$$

- γ is given by $\gamma = 7 \left(\frac{gF}{W^2} \right)^{-0.143}$

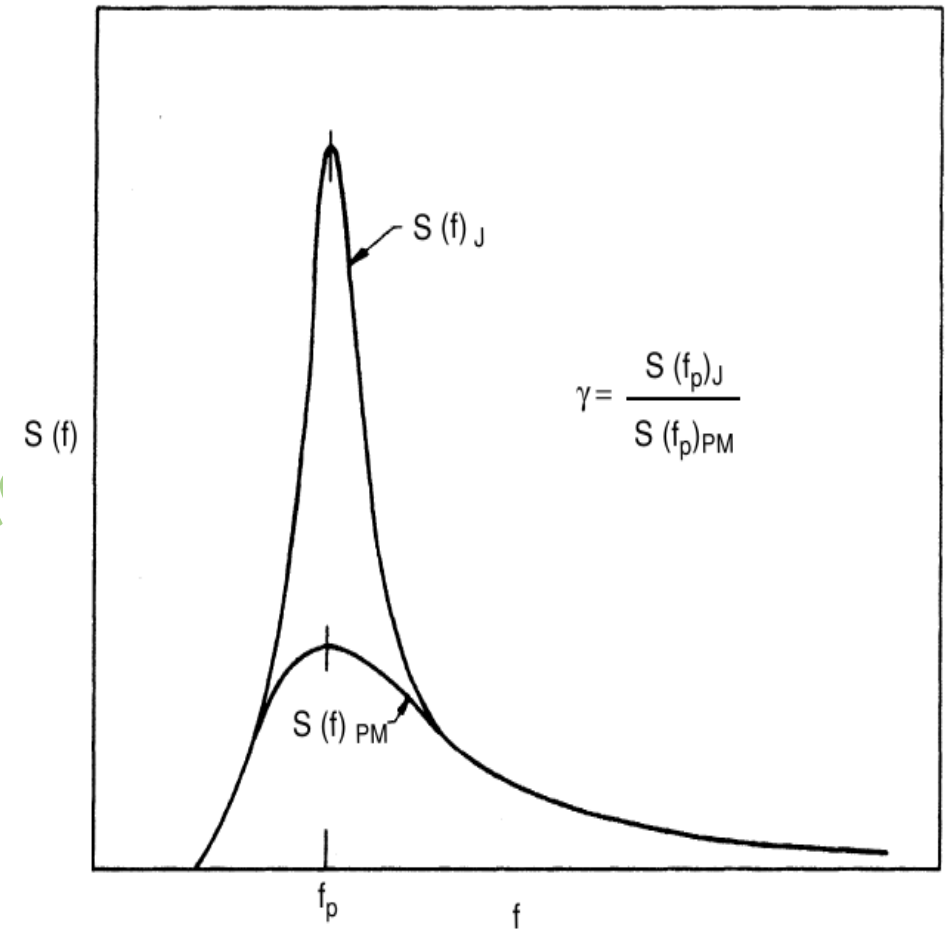


Figure 6. JONSWAP and PM spectra comparison.

Wave Spectral Models

TMA Spectrum (Bouws et al., 1985)

- The previous three models were developed for deep water conditions.
- As wind waves propagate into intermediate and shallow depths there is a period-dependent change in the shape of the spectrum versus that for deep water.
- The TMA spectrum is a wave spectrum based on the generation of waves in deep water that then propagate without refracting into intermediate/shallow water depths.
- The spectral form is a JONSWAP spectrum modified by a depth and frequency dependent factor $\Phi(f, d)$. Thus, $S(f)_{\text{TMA}} = S(f)_J \Phi(f, d)$ where $\Phi(f, d)$ is a relatively complex function defined graphically in Figure 7.

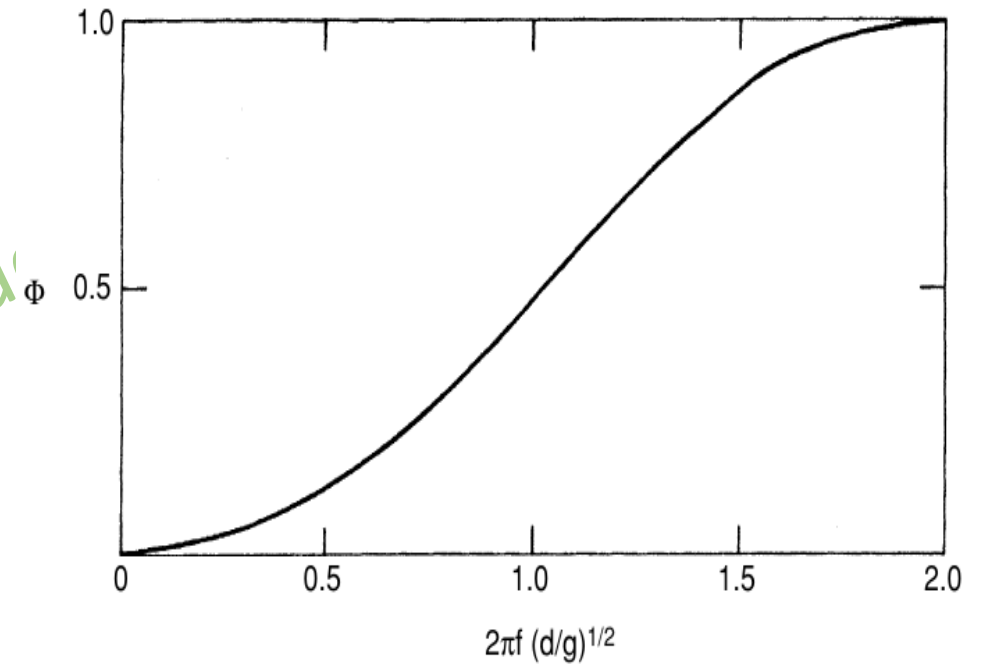


Figure 7. Correction factor for TMA spectrum.

Wave Spectral Models

- Hughes (1984) further proposed that α and γ in the JONSWAP spectral formulation be modified to

$$\alpha = 0.0078 \left(\frac{2\pi W^2}{gL_p} \right)^{0.49}$$

- For the TMA spectrum. L_p is the wave length

$$\gamma = 2.47 \left(\frac{2\pi W^2}{gL_p} \right)^{0.39}$$

- For the spectral peak frequency and the water depth for which the TMA spectrum is being determined.

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Wave Prediction - SMB method

- Consider a dimensional analysis of the basic deep water wave prediction relationship

$$H_s, T_s = f(W, F, t_d, g)$$

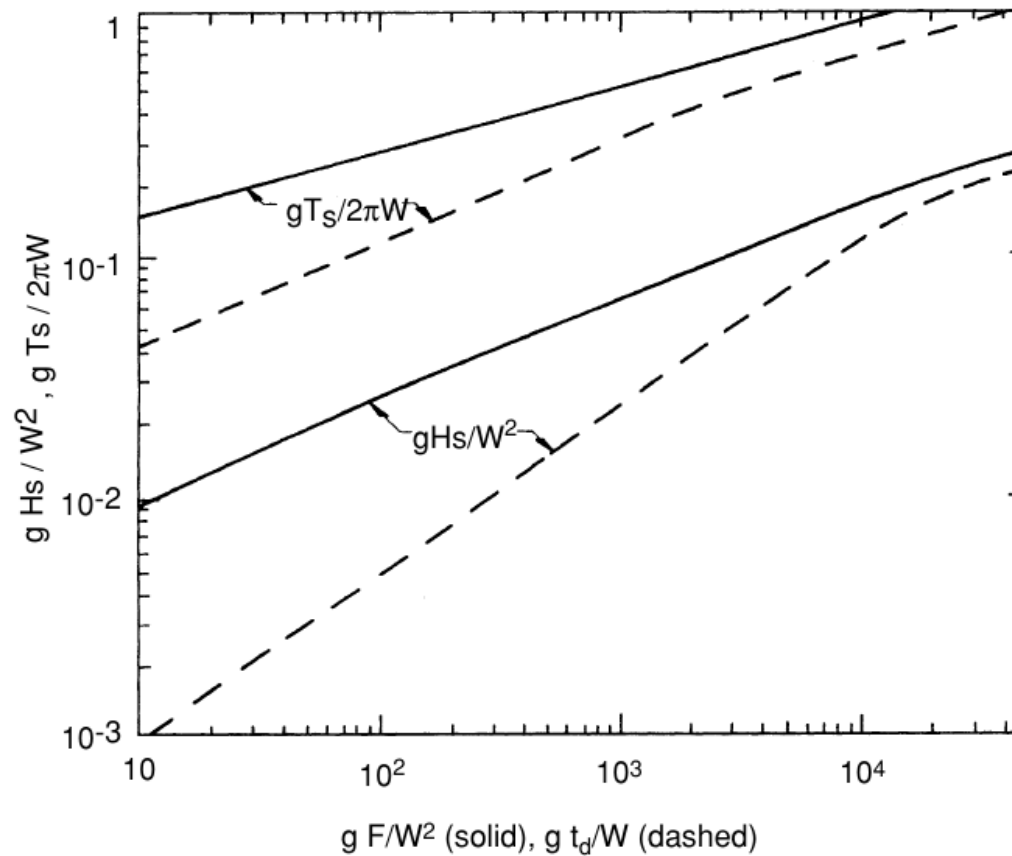
which leads to

$$\frac{gH_s}{W^2}, \frac{gT_s}{2\pi W} = f\left(\frac{gF}{W^2}, \frac{gt_d}{W}\right)$$

- The above simply relates the dimensionless significant wave height and period to the dimensionless fetch and duration. Either the fetch or the duration term on the right would control, depending on whether wave generation were fetch or duration limited.
- Also, since $C = gT/2\pi$ for deep water waves, the second term on the lefthand side can be written C/W , a parameter known as the wave age and important to the understanding of wind wave growth.

Wave Prediction - SMB method

- Figure 8 presents the relationship as a dimensionless plot as given by $\frac{gH_s}{W^2}, \frac{gT_s}{2\pi W} = f\left(\frac{gF}{W^2}, \frac{gt_d}{W}\right)$



A deep lake has a wind with an average velocity of 30 m/s blowing over it for a period of 2 hours. The fetch in the direction of the wind is 20 km. Using the SMB method, what significant wave height and period will be generated at the downwind end of the lake after two hours?

Solution:

For the given fetch and wind speed

$$\frac{gF}{W^2} = \frac{9.81(20,000)}{(30)^2} = 218$$

Figure 8 then yields

$$\frac{gH_s}{W^2} = 0.034$$

$$\frac{gT_s}{2\pi W} = 0.33$$

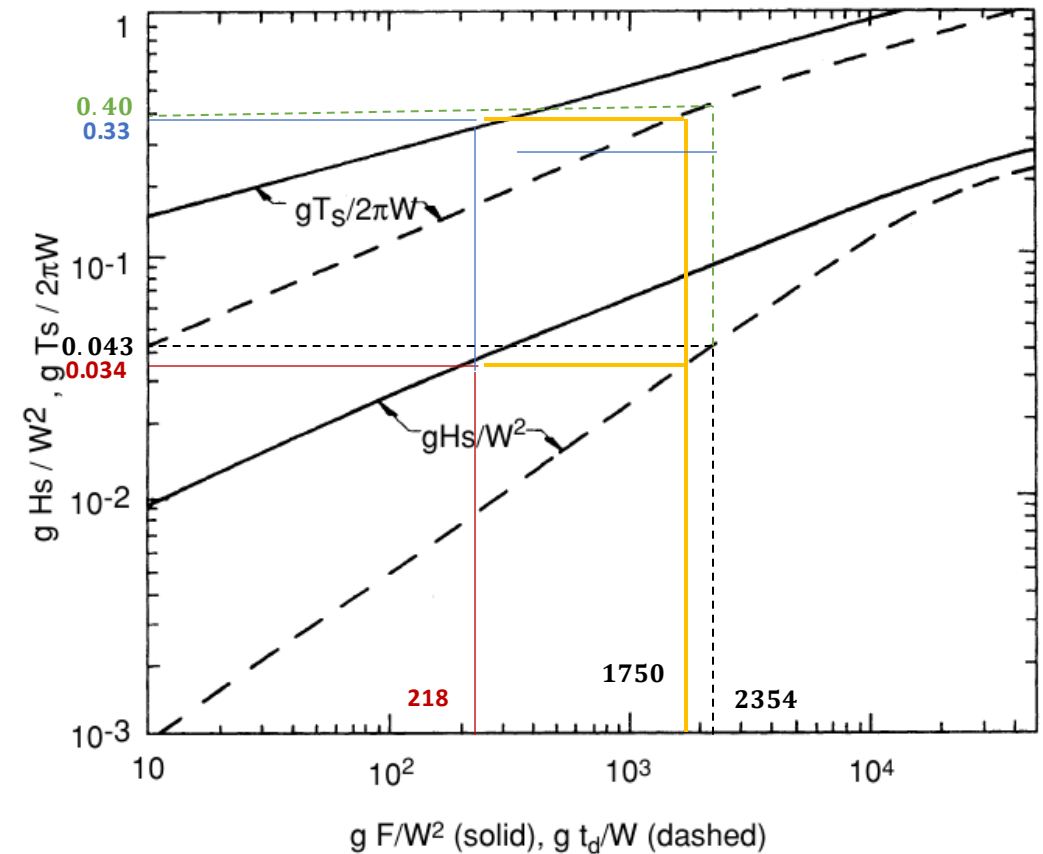


Figure 8. SMB wave prediction curves.

or

$$H_s = \frac{0.034(30)^2}{9.81} = 3.1 \text{ m}$$
$$T_s = \frac{0.33(2)\pi(30)}{9.81} = 6.4 \text{ s}$$

For the given duration and wind speed

$$\frac{gt_d}{W} = \frac{9.81(2)(3600)}{30} = 2354$$

Figure 8 then yields

$$\frac{gH_s}{W^2} = 0.043 \quad \frac{gT_s}{2\pi W} = 0.40$$

or

$$H_s = \frac{0.043(30)^2}{9.81} = 3.9 \text{ m}$$
$$T_s = \frac{0.40(2)\pi(30)}{9.81} = 7.7 \text{ s}$$

The smaller values, $H_s = 3.1$ m and $T_s = 6.4$ s control and wave generation is fetch limited.

Note, that for

$$\frac{gH_s}{W^2} = 0.034 \quad \frac{gT_s}{2\pi W} = 0.33$$

Figure 8 yields

$$\frac{gt_d}{W} = 1750$$

or

$$t_d = \frac{1750(30)}{9.81} = 5350 \text{ s (1.5 hours)}$$

- Thus, after the wind has blown for about 1.5 hours the waves at the downwind end of the lake reach their limiting height and period. During the remaining half hour of wind, wave conditions would remain about the same.

Wave Prediction—Spectral Models

SPM-JONSWAP procedure

- To apply this wave prediction procedure, first determine the adjusted wind speed W_A given by

$$W_A = 0.71W^{1.23}$$

where W_A and W are both in meters per second.

- Then the significant height and peak period can be calculated from

$$\begin{aligned}\frac{gH_{mo}}{W_A^2} &= 0.0016 \left(\frac{gF}{W_A^2} \right)^{0.5} \\ \frac{gT_p}{W_A} &= 0.286 \left(\frac{gF}{W_A^2} \right)^{0.33}\end{aligned}$$

Wave Prediction—Spectral Models

- The values determined from the above two equations use only the wind speed and fetch and are thus only for the fetch-limited condition. A limiting wind duration would be calculated from

$$\frac{gt_d}{W_A} = 68.8 \left(\frac{gF}{W_A^2} \right)^{0.66}$$

- If the actual duration is greater than the duration calculated the wind generation process is fetch limited and the results are the predicted significant height and peak period.
- If the actual duration is less, the process is duration limited.

Problem

A deep lake has a wind with an average velocity of 30 m/s blowing over it for a period of 2 hours. The fetch in the direction of the wind is 20 km., calculate the significant height and peak period using the SPM-JONSWAP procedure.

$$\text{Use, } W_A = 0.71W^{1.23}$$

$$\frac{gH_{mo}}{W_A^2} = 0.0016 \left(\frac{gF}{W_A^2} \right)^{0.5}$$

$$\frac{gT_p}{W_A} = 0.286 \left(\frac{gF}{W_A^2} \right)^{0.33}$$

$$\frac{gt_d}{W_A} = 68.8 \left(\frac{gF}{W_A^2} \right)^{0.66}$$

Problem

Solution:

The adjusted wind speed is

$$W_A = 0.71(30)^{1.23} = 46.6 \text{ m/s}$$

$$H_{mo} = 0.0016 \left(\frac{9.81(20,000)}{(46.6)^2} \right)^{0.5} \frac{(46.6)^2}{9.81} = 3.4 \text{ m}$$

$$T_p = 0.286 \left(\frac{9.81(20,000)}{(46.6)^2} \right)^{0.33} \left(\frac{46.6}{9.81} \right) = 6.1 \text{ s}$$

$$t_d = 68.8 \left(\frac{9.81(20,000)}{(46.6)^2} \right)^{0.66} \left(\frac{46.6}{9.81} \right) = 6580 \text{ s (1.82 hours)}$$

Since the actual duration is greater than the calculated duration, the wave generation process is fetch limited and $H_{mo} = 3.4 \text{ m}$, $T_p = 6.1 \text{ s}$.

These values are close to the values $H_s = 3.1 \text{ m}$ and $T_s = 6.4 \text{ s}$ given by the SMB method and the process is fetch limited as indicated by that method.

Long-term Analysis

- In the preceding sections, methods related to analysis of data pertaining to waves that occur in one or few wave trains, that too measured over a short period, were dealt.
- In order assess most probable wave height that might strike a coastal or an offshore facility to its life time, long-term analysis is carried out.
- For this purpose, wave data recorded over a long period would be necessary. Procedure adopted to determine wave height with ' N ' year return period involves
 - (i) obtaining data on wave heights observed over a long period
 - (ii) selecting a probability distribution and
 - (iii) fitting wave data and extrapolate to determine the extreme wave height.

Long-term Analysis

Commonly adopted distributions are

- (i) Log-normal distribution,
- (ii) Gamma distribution
- (iii) Weibull distribution
- (iv) Fisher-Tippett type I or Gumbel distribution.

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Long-term Analysis

- **Log-normal distribution :**

Two parameter log-normal distribution is given by

$$f(x) = \frac{1}{x\beta\sqrt{2\pi}} \exp \left[- \left(\frac{[\ln x - \alpha]^2}{2\beta^2} \right) \right], x > 0$$

$$f(x) = 0, \text{ otherwise}$$

With

$$\text{mean} = \exp \left[\alpha + \frac{\beta^2}{2} \right]$$

$$\text{variance} = \exp[2\alpha] \exp[\beta^2] [\exp(\beta^2) - 1]$$

Long-term Analysis

- **Gamma distribution**

$$f(x) = \frac{x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} \exp\left[-\frac{x}{\beta}\right], \alpha, \beta, x > 0$$
$$= 0, \text{ otherwise}$$

with

$$\begin{aligned} \text{mean} &= \alpha\beta \\ \text{variance} &= \alpha\beta^2 \end{aligned}$$

- **Two parameter Weibull distribution**

$$f(x) = \beta \left(\frac{x}{\alpha}\right)^{\beta-1} \frac{1}{\alpha} \exp\left[-\left(\frac{x}{\alpha}\right)^\beta\right], \alpha, \beta, x > 0$$
$$= 0, \text{ otherwise}$$

$$\text{mean} = \alpha \Gamma\left[\frac{1}{\beta} + 1\right]$$

$$\text{variance} = \alpha^2 \Gamma\left(\frac{2}{\beta} + 1\right) - \mu^2$$

Long-term Analysis

- **Fisher-Tippett Type I (Gumbel distribution)**

$$f(x) = \frac{1}{\beta} \exp \left[- \left(\frac{x - \alpha}{\beta} \right) - \exp \left(- \frac{x - \alpha}{\beta} \right) \right], \beta > 0$$

with

$$\text{mean} = \alpha + \gamma\beta$$

$$\text{variance} = \beta^2 \frac{\pi^2}{6}$$

$$\gamma \text{ Euler's constant} = 0.5772$$

- **Three parameter Weibull distribution**

$$f(x) = \frac{\beta}{\alpha} \left(\frac{x - \theta}{\alpha} \right)^{\beta-1} \exp \left[- \left(\frac{x - \theta}{\alpha} \right)^{\beta} \right]$$

where,

β = the shape parameter,

α = the scale parameter and

θ = the location parameter.

Long-term analysis based on Gumbel Distribution

- Gumbel probability density and cumulative distribution functions for wave height are given as
Where

$$f(H) = \frac{1}{\beta} \exp \left[- \left(\frac{H - \alpha}{\beta} \right) - \exp \left(- \frac{H - \alpha}{\beta} \right) \right], \beta > 0$$

$$F(H) = \exp \left[- \exp \left(- \frac{H - \alpha}{\beta} \right) \right]$$

$$F(H) = F(H^* \leq H)$$

where H^* is any given wave height and

H specified wave height

$Q(H) = 1 - F(H)$, the probability of exceedance.

Taking logs on both the sides, we have

$$\ln F = - \exp \left[- \frac{H - \alpha}{\beta} \right]$$

Long-term analysis based on Gumbel Distribution

$$-\ln[-\ln F] = \frac{H - \alpha}{\beta}$$

- In the form

$$Y = mX + C$$

$$-\ln \left[\ln \frac{1}{F} \right] = \frac{1}{\beta} H + \left[-\frac{\alpha}{\beta} \right]$$

with Slope $m = \frac{1}{\beta},$

$$X = H,$$

Intercept $C = -\frac{\alpha}{\beta}$ and

$$Y = -\ln \left[\ln \frac{1}{F} \right]$$

Long-term analysis based on Gumbel Distribution

The wave height H with a return period of T_R years for Gumbel distribution is given as

$$H_{T_r} = \alpha - \beta \ln \left[\ln \left\{ \frac{N_y T_r}{N_y T_r - 1} \right\} \right]$$

where

N_y = Number of events per year

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Long-term analysis based on Gumbel Distribution

- Let us assume that information on hourly wave heights obtained through wave hindcasting techniques for a period of 40 years is available for a station.
- Figure 9 shows one such case wherein with 3,50,400 wave heights, an analysis is carried out to determine the number of peak values exceeding an arbitrarily chosen threshold value of $H = 1.5$ m.
- From the figure, probability that the number of waves less than 2 m is found by dividing $(6492 - 4770)$ by 6492, i.e. $F(2) = 0.265$. Similarly, for wave height equal to 2.5, $F(2.5) = (6492 - 3090)/6492 = 0.524$, and so on for other wave heights from 3 to 5.5.

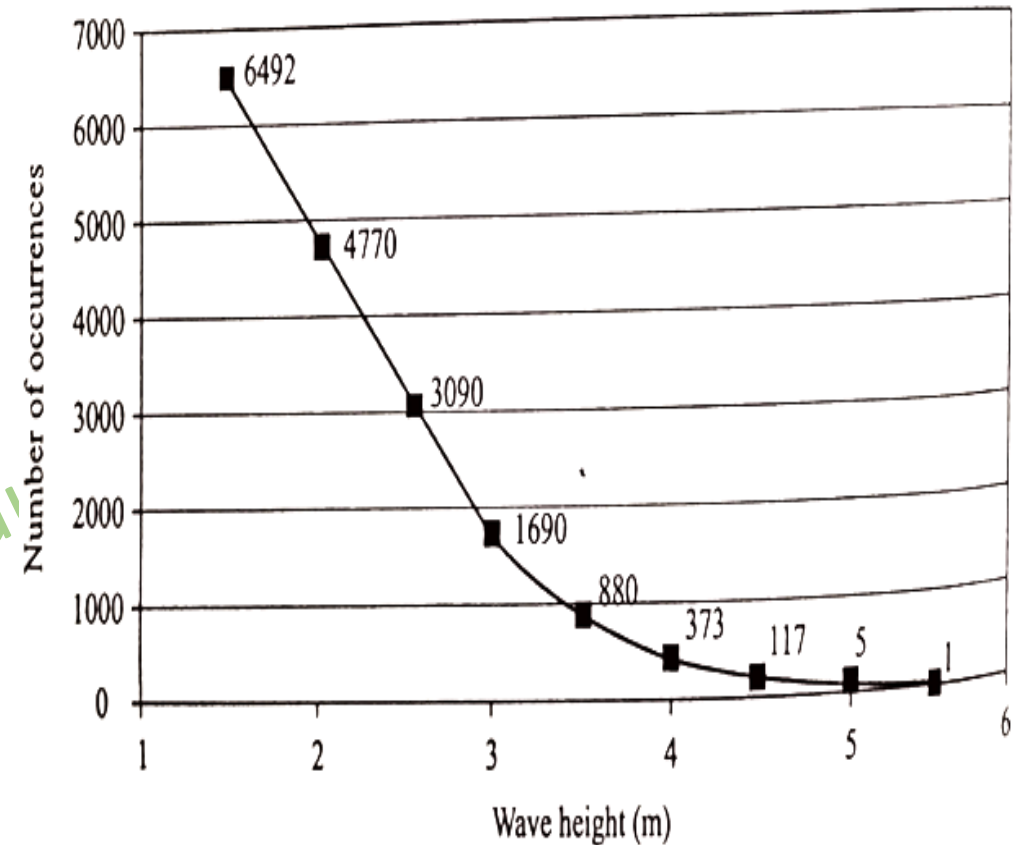


Figure 9. Number of occurrences with the given threshold wave height

Long-term analysis based on Gumbel Distribution

- A plot of wave height with $Y = -\ln(\ln(1/F))$ shows that the data can be fitted with a straight line with slope $m = 2.40$ and intercept $C = -5.89$ (Figure 10).
- With $\beta = 1/m$ and $\alpha = -(C \times \beta)$, β and α are determined as 0.417 and 2.46, respectively.

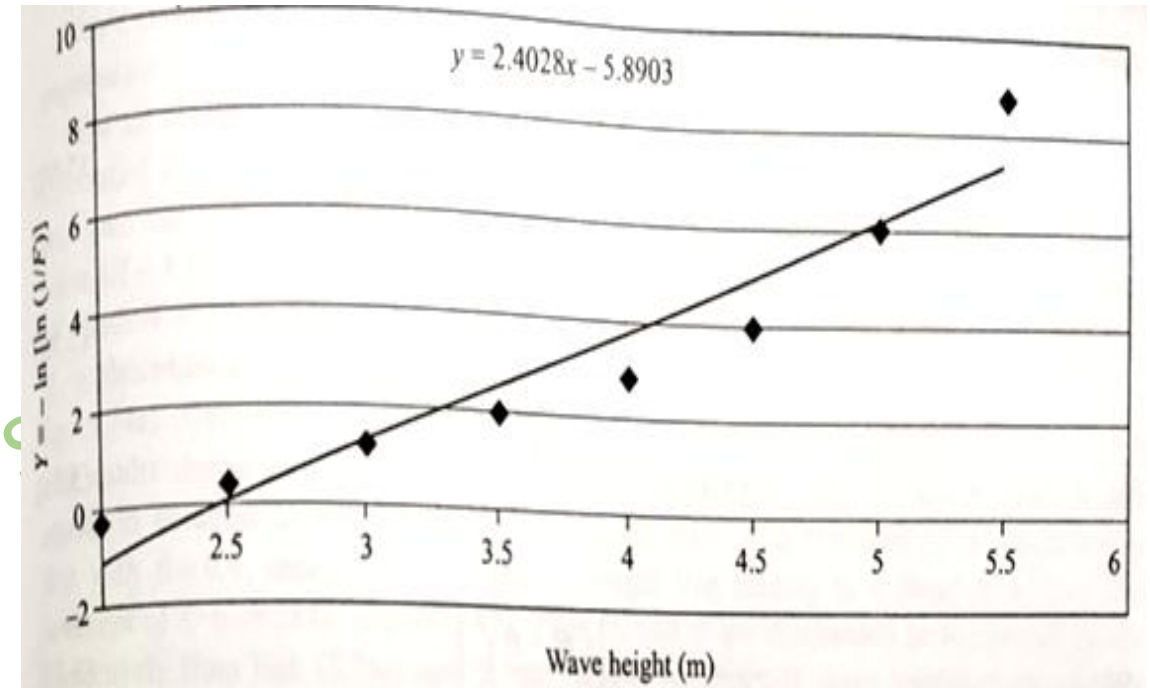


Figure 10. Plot based on Gumbel distribution with threshold wave height $H = 1.5$ m.

Long-term analysis based on Gumbel Distribution

- Now to determine the extreme wave height when analysis is carried out with Gumbel distribution.
- For example, for a wave height of 1.5 m, $N_y (= 6492/40)$ is 162.
- Thus, a wave with return period of 50 years would have a height of

$$H_{T_r} = 2.46 - 0.417 \ln \left\{ \ln \left[\frac{162 \times 50}{8099} \right] \right\}$$
$$H_{T_r} = 6.21 \text{ m}$$

- The above exercise can be repeated for other threshold values and extreme wave heights determined for given T_r .